ST MARY'S DSG, KLOOF

GRADE 10 MATHEMATICS

Prop Int Theorem, Inequalities, Number Systems and Straight lines

May 2023

TIME: 80 MINUTES **TOTAL: 66 MARKS**

NAME: _____ **MATHS TEACHER:_____**

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READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

1. This question paper consists of 9 pages and 7 questions.
2. Answer all the questions on the question paper.
3. All the necessary working details must be clearly shown.
4. Approved non-programmable calculators may be used except where otherwise stated.
5. It is in your own interest to write legibly and to present your work neatly.

QUESTION 1

Given the numbers :

$$\mathbf{3^3}\ ;\ -\sqrt[3]{\mathbf{27}}\ ;\ \mathbf{30}\ ;\ \mathbf{3\pi}\ ;\ \sqrt{\frac{\mathbf{4}}{\mathbf{25}}}\ ;\ \sqrt{\mathbf{3}}\ ;\ \sqrt{\mathbf{-4}}\ ;\ \mathbf{-2,\dot{1}\dot{4}}\ ;\ (\sqrt{\mathbf{3}})^{\mathbf{2}}$$

a) Which of these numbers are :

1. An integer with factors 2 and 5.

2. Rational numbers that are not integers.

3. An integer, but not a counting number.

4. Non real number/s

5. All the irrational numbers.

(5)

b) Given : $A = \sqrt{4\,x - 3}$

Give ONE value of x that satisfies the condition for A.

1. $A = 0$ (1)

2. Integer (2)

3. Irrational (2)

4. Non-real (2)

c) The value of $\sqrt{33}$ lies between 2 integers. Find these integers without a calculator.

(2)

[14]

QUESTION 2

The following inequality is given :

$\boldsymbol{-9 \leq 2x + 3 < 5\ ,\quad x \epsilon R}$

a) Solve for x. (2)

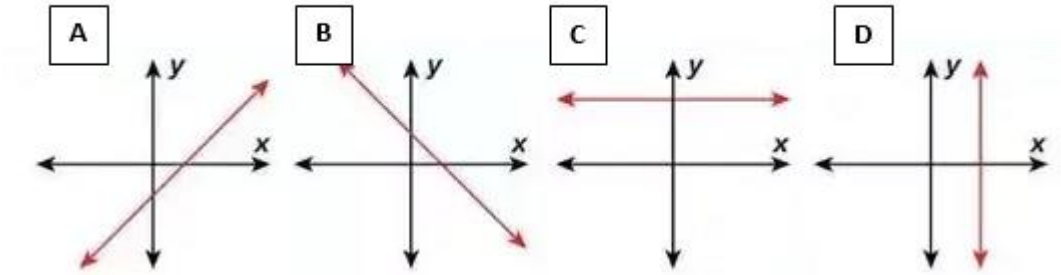
b) Represent your answer on the number line. (2)

c) Write your answer in Interval Notation. (2)

[6]

QUESTION 3

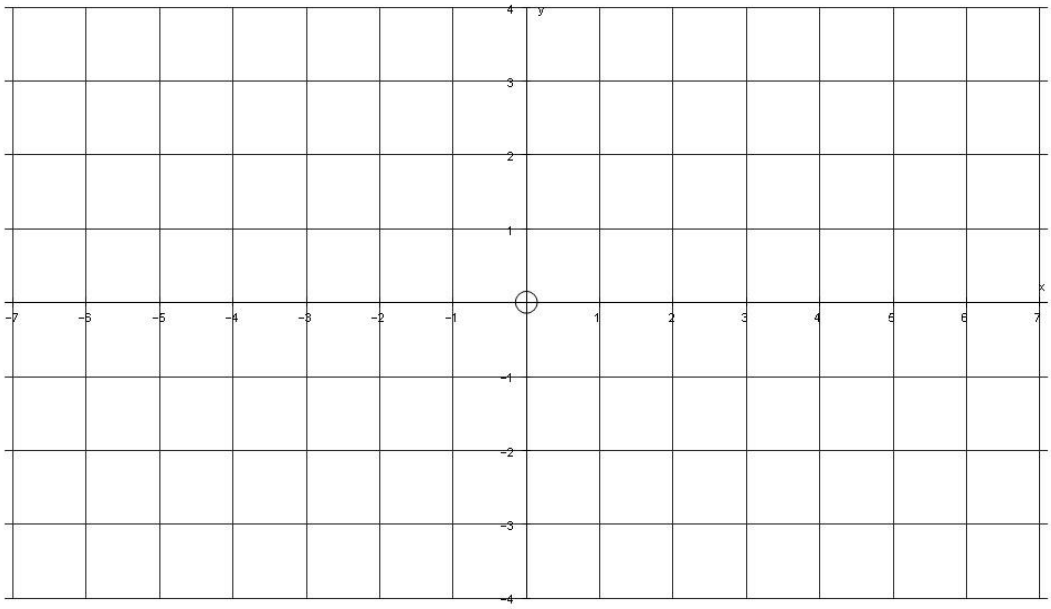
Write down, below each graph, if the gradient is positive , negative, equal to zero or undefined.



[4]

QUESTION 4

a) Sketch the graphs of $y = -2x - 2$ and $3y - 6x = 9$ on the set of axes below and answer the questions that follow.



Label your graphs and the intercepts. (6)

b) What is the equation of the y-axis?

(1)

c) Label the point of intersection between the above graphs with the letter A, and write down the coordinates of the point of intersection. (2)

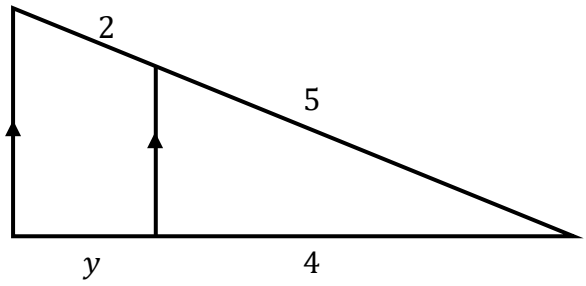
d) Determine the equation of a line that is parallel to $y = -2x - 2$ and has a y-intercept of 1. (2)

e) Determine the equation of a line that passes through the points (2;5) and (-2;1) (3)

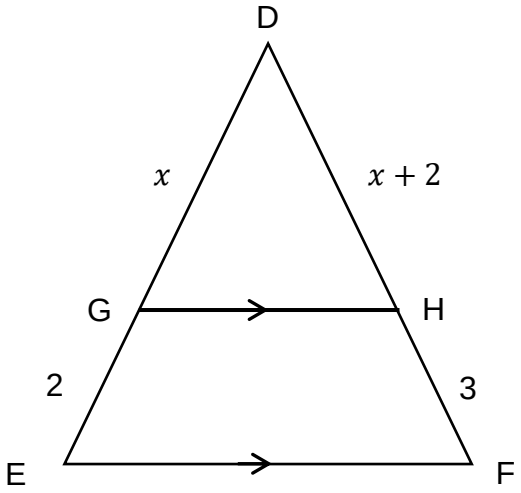
[14]

QUESTION 5

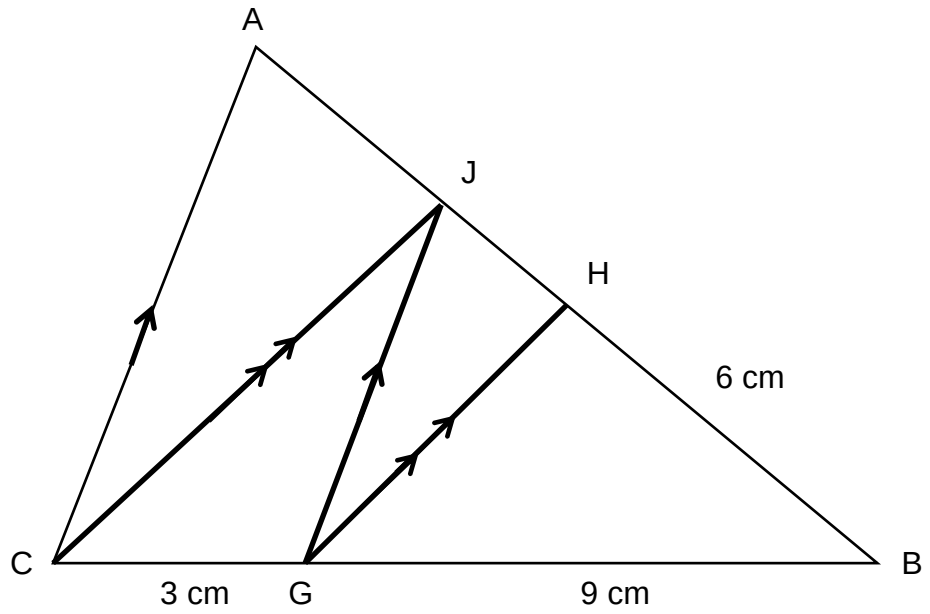
a) Find the value of y, giving reasons. (2)



b) Find the value of x, giving reasons. (3)



c) Given triangle ABC with AC // JG, JC // HG and GB is 9 cm, CG = 3cm and BH = 6cm.



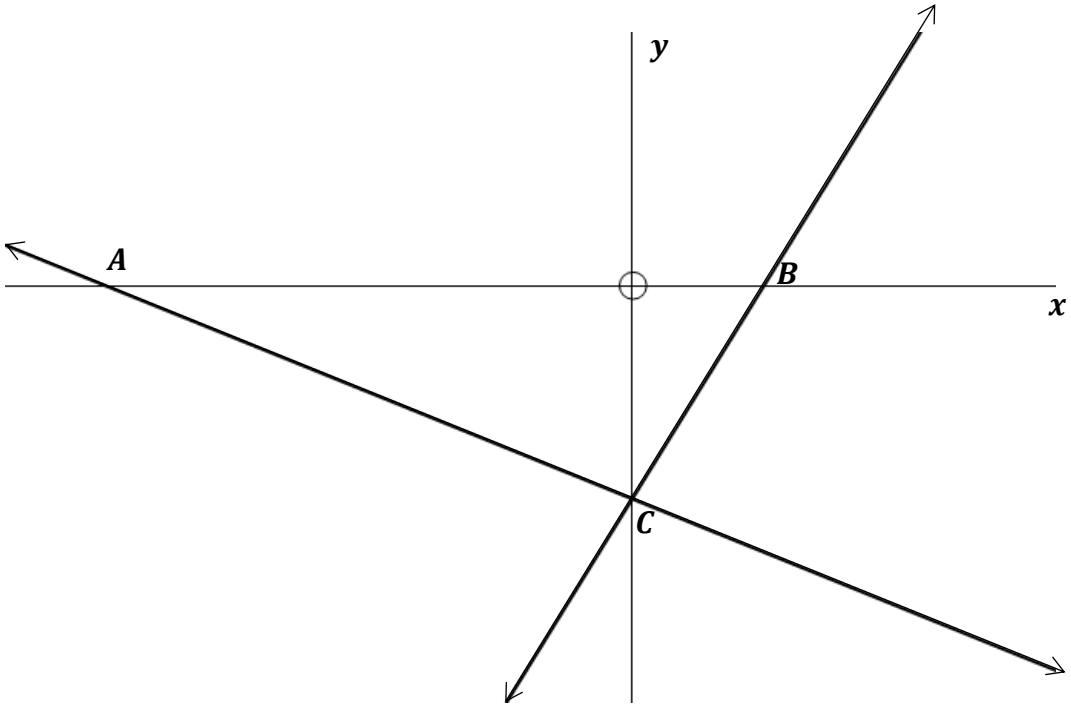
1. Find JH, giving reasons. (3)

2. Find AJ, giving reasons. (3)

[11]

QUESTION 6

There are 2 graphs $y = = 2x - 3$ and $y = -\frac{1}{2}x - 3$.



a) Label the graphs on the Cartesian plane correctly (2)

b) Determine $A\hat{C}B$, giving a reason. (2)

c) Determine the length of AB (3)

d) Find the area of ΔABC. (3)

[10]

QUESTION 7

Show all working details

a) An irrational number that lies between 5 and 6 is : (2)

1) $\sqrt[3]{28}$
2) $\sqrt{20+8}$
3) $\sqrt{20\times 8}$
4) 2π

b) The formula for the volume of the cone is $V=\frac{1}{3}\pi r^2 h$. The radius, r, of the cone may be expressed as : (2)

1) $\sqrt{\frac{3V}{\pi h}}$
2) $\sqrt{\frac{V}{3\pi h}}$
3) $\sqrt[3]{\frac{V}{\pi h}}$
4) $\frac{1}{3}\sqrt{\frac{V}{\pi h}}$

c) If k is a negative single digit integer, which of the following is **NOT** a possible value of $\mathbf{2-k}$? (3)

1) 4
2) 8
3) 10
4) 12

[7]